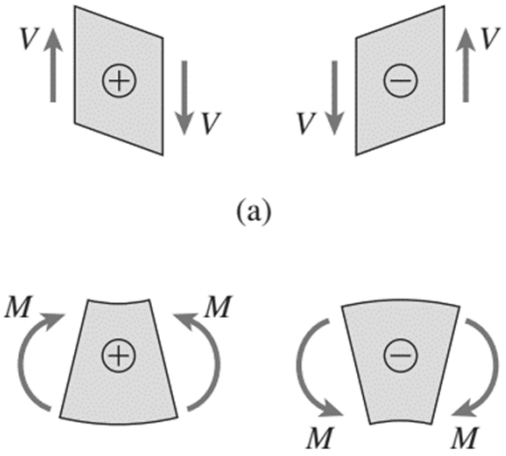
Shear Force and Bending Moments

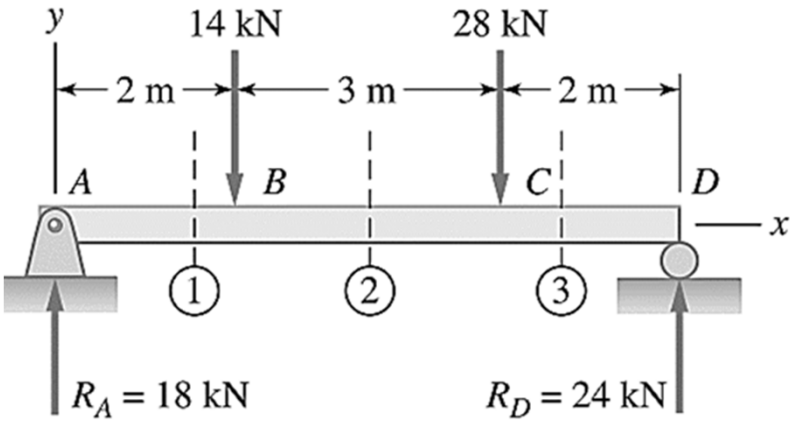


Shear Force and Bending Moment equations approach

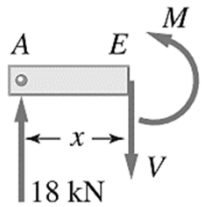
➢ Procedure

- ✓ Compute the support reactions from the FBD of the entire beam.
- ✓ Divide the beam into sections (segments) so that, the end-points of the sections are discontinuities of loading, which include including concentrated loads and couples.
- ✓ Introduce an imaginary cutting plane within the segment, located at a distance x from the left end of the beam.
- ✓ Draw a FBD for a convenient part of the beam and shown show V and M acting in their positive directions.
- ✓ Determine the expressions for V and M from the equilibrium equations obtainable from the FBD. These expressions, which are usually functions of x, are the shear force and bending moment equations for the segment.
- ✓ Plot the expressions for V and M for the segment.

Shear Force and Bending Moments



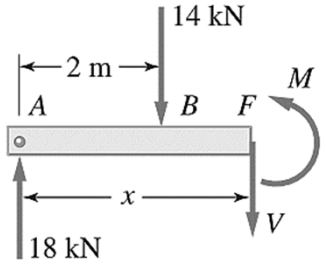
For segment AB ($0<x<2$)



$V = 18$ kN

$M_E = 18x$ kN·m

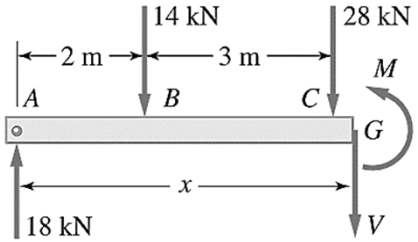
For segment BC ($2<x<5$)



$V = 4$ kN

$M_F = (4x + 28)$ kN·m

For segment BC ($5<x<7$)



$V = -24$ kN

$M_G = (-24x + 168)$ kN·m

Shear Force and Bending Moments

For segment AB ($0<x<2$)

$V = 18$ kN

$M = 18x$ kN·m

For segment BC ($2<x<5$)

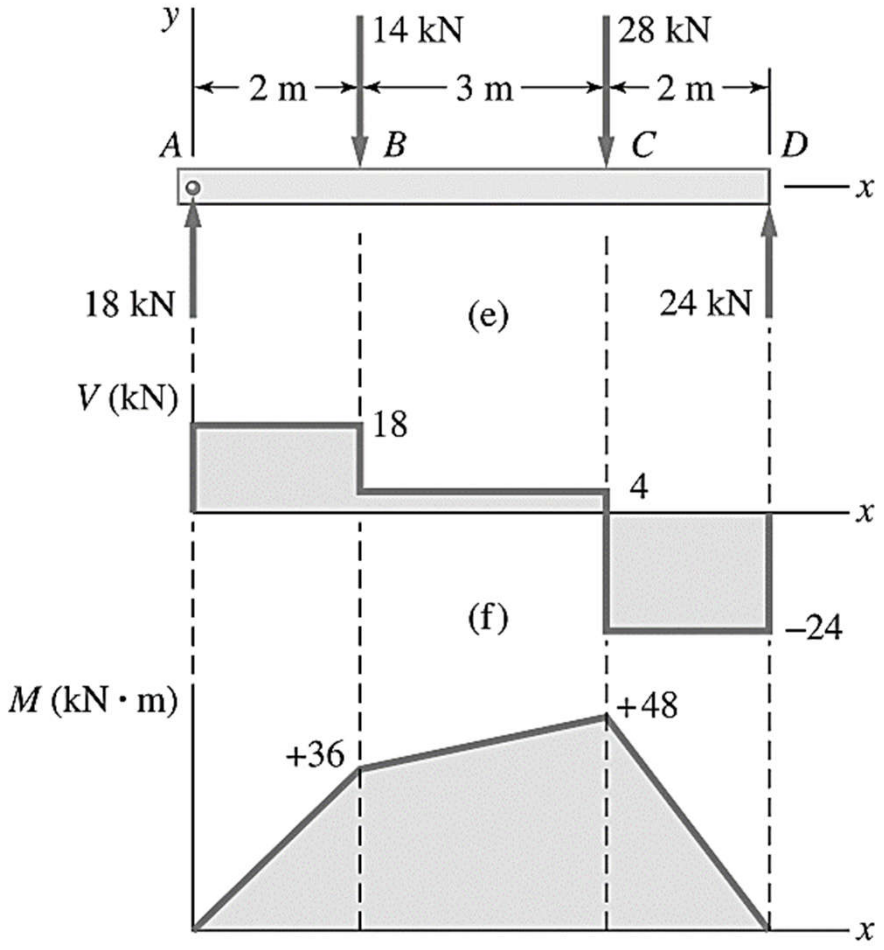
$V = 4$ kN

$M = (4x + 28)$ kN·m

For segment BC ($5<x<7$)

$V = -24$ kN

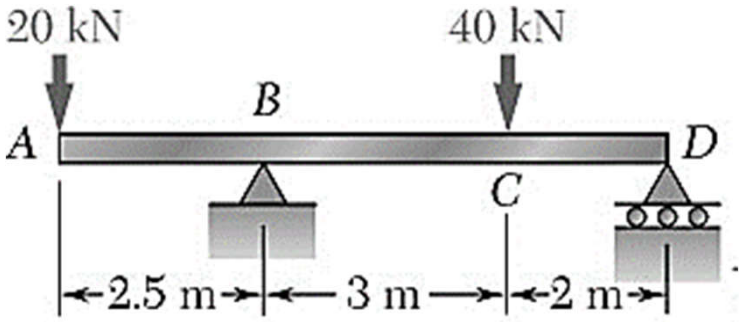
$M = (-24x + 168)$ kN·m

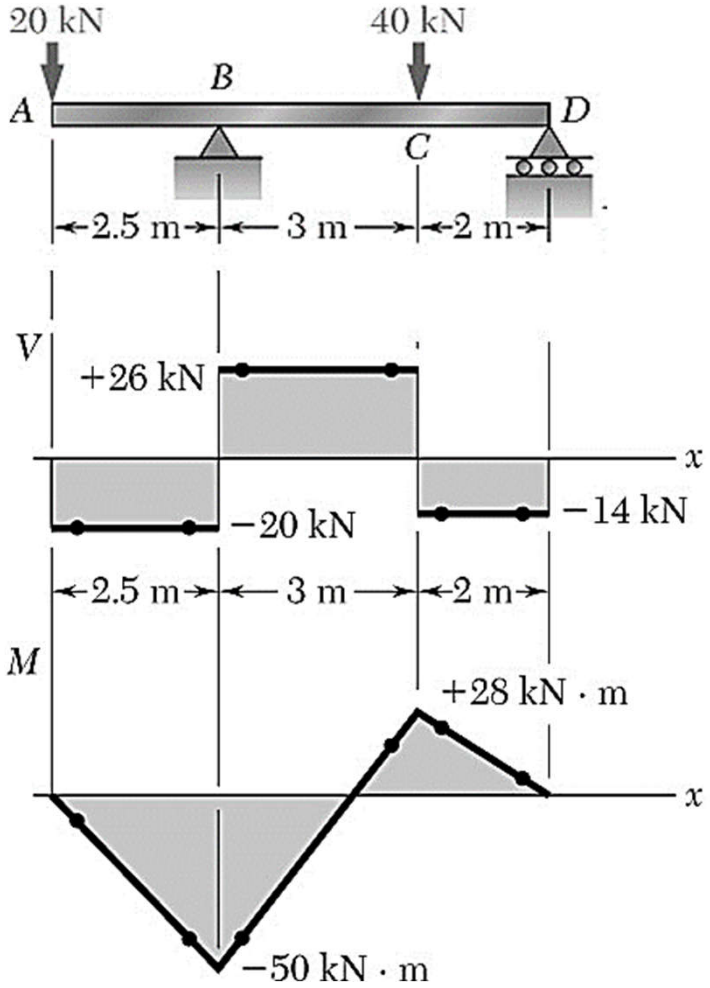


V diagram

M diagram

Shear Force and Bending Moments

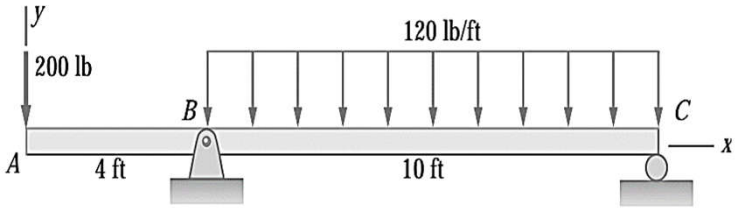


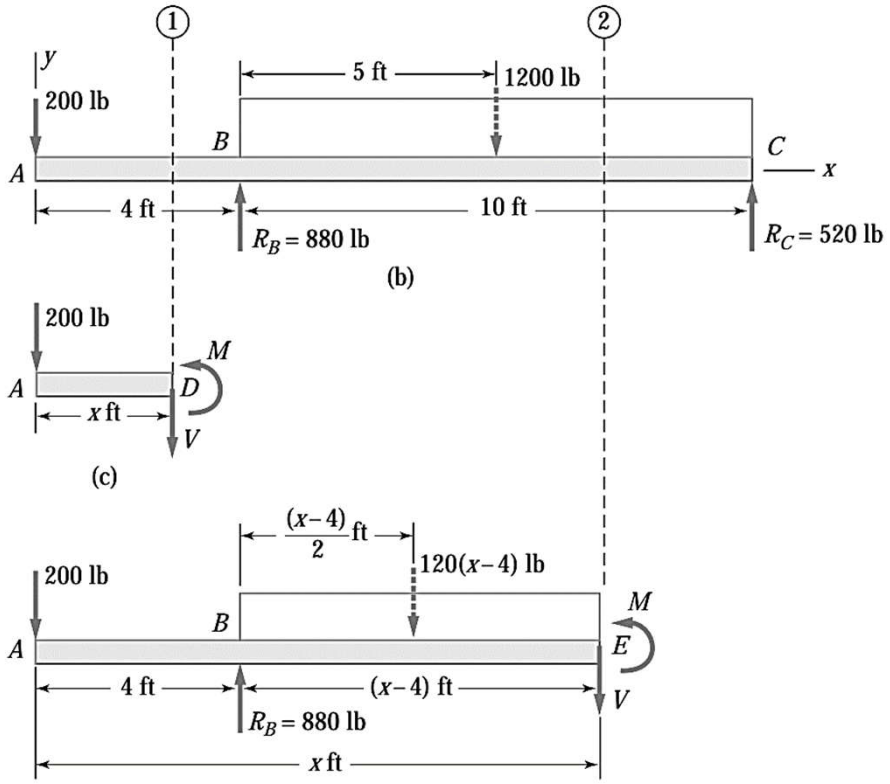


Shear Force and Bending Moments

Distributed Loads (constant intensity)

Replace distributed load with a point load equal to the maximum load (kinda like the area formed by the profile formed by the distributed load). Point load should be at centroid of profile (kinda like the area formed by the distributed load, so to speak)





For segment AB ($0<x<4$)

$V = -200$ lb

$M_D = -200x$ lb·ft

For segment BC ($4<x<14$)

$V = (1160 - 120x)$ lb

$M_E = (60x^2 + 1160x + 4480)$ lb·ft

Shear Force and Bending Moments

For segment AB ($0<x<4$)

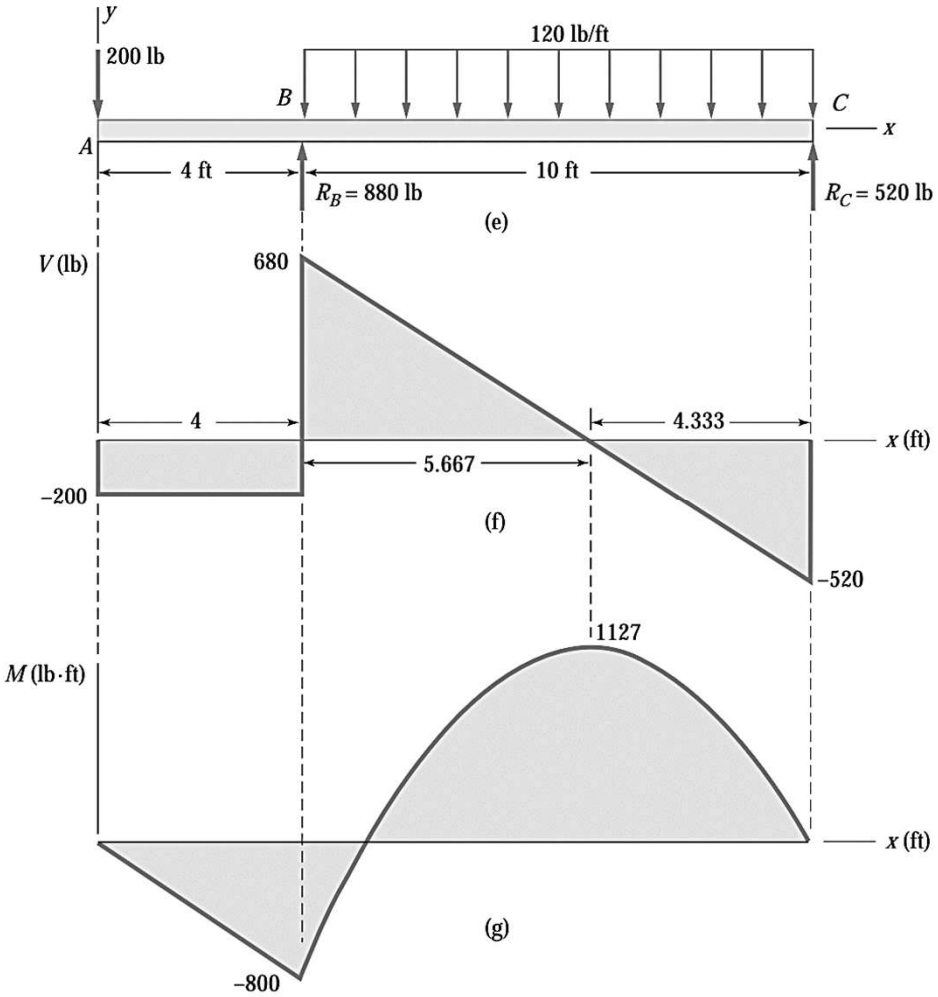
$V = -200$ lb

$M = -200x$ lb·ft

For segment BC ($4<x<14$)

$V = (1160 - 120x)$ lb

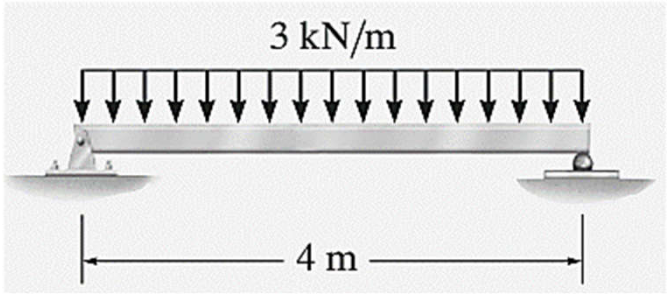
$M = (60x^2 + 1160x + 4480)$ lb·ft



V diagram

M diagram

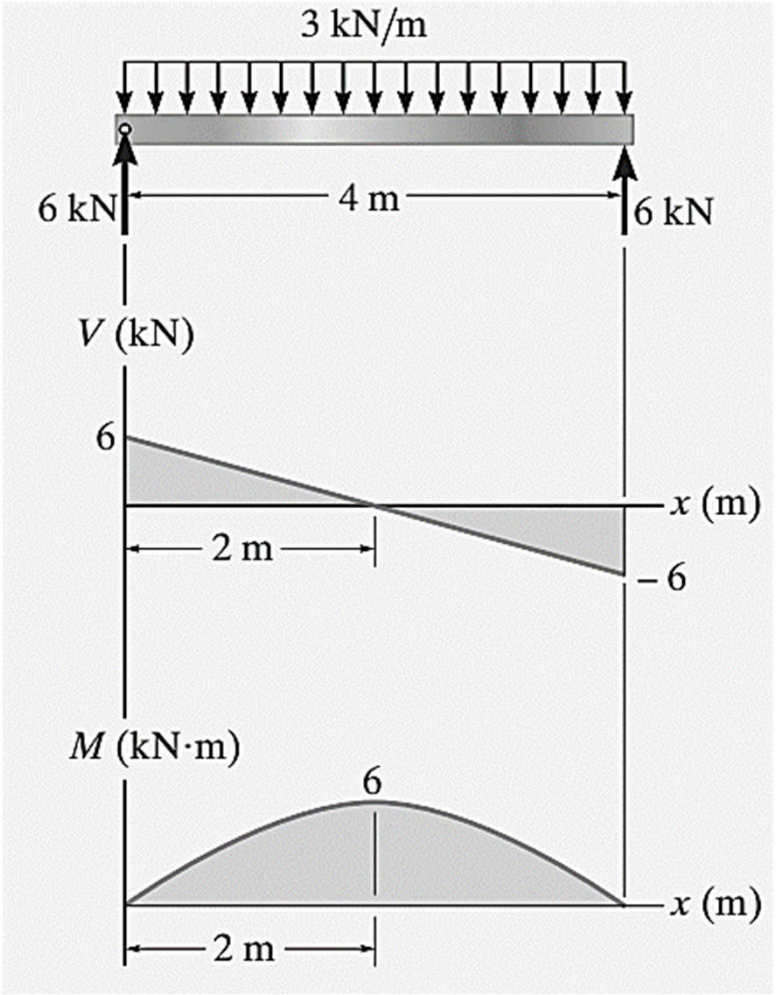
Shear Force and Bending Moments



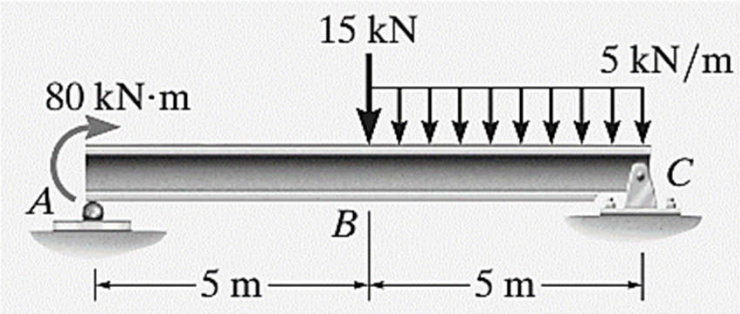
For segment ($0<x<4$)

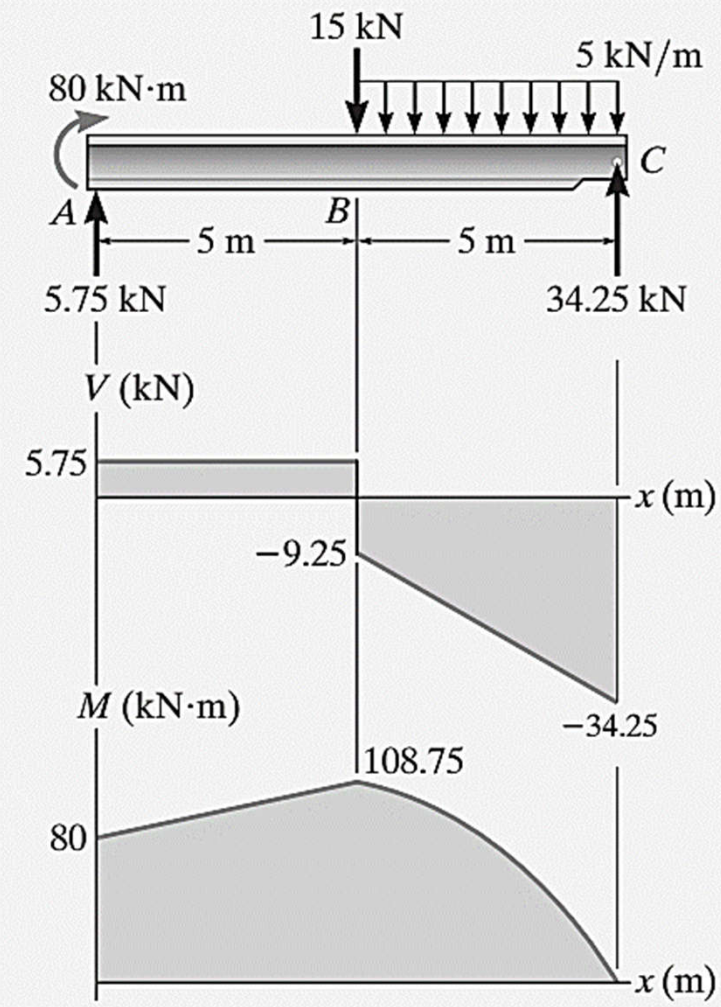
$V = (6-3x)$ kN

$M = (6x-1.5x^2)$ kN·m



Shear Force and Bending Moments



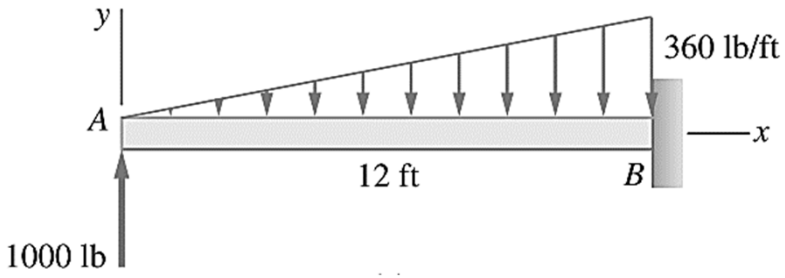


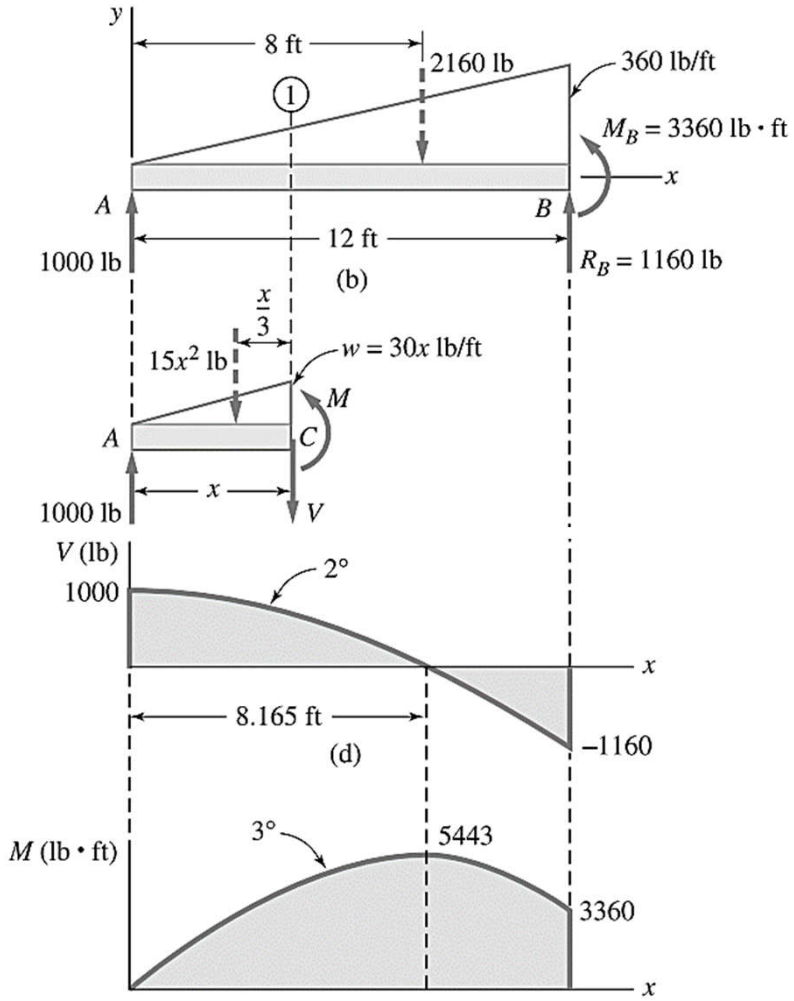
Shear Force and Bending Moments

Distributed Loads (varying intensity)

Replace distributed load with a point load equal to the maximum load (kinda like the area formed by the profile of the distributed load). Point load should be at centroid of area formed by the profile (so to speak).

For section FBDs, replace distributed load with load intensity, calculated from ratio; (intensity/x) = (point load/distance of distributed load)



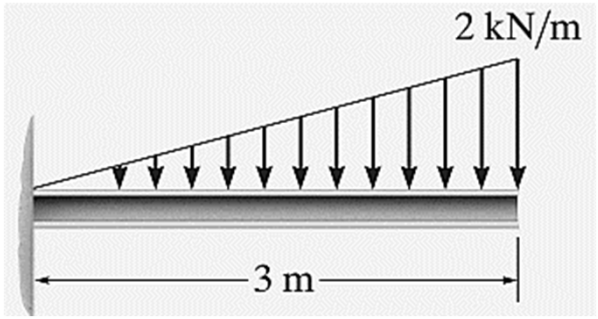


For segment AB (*0<x<12*)

$$V = 1000 - 15x^2 \text{ lb}$$

$$M_C = 1000x - 5x^3 \text{ lb} \cdot \text{ft}$$

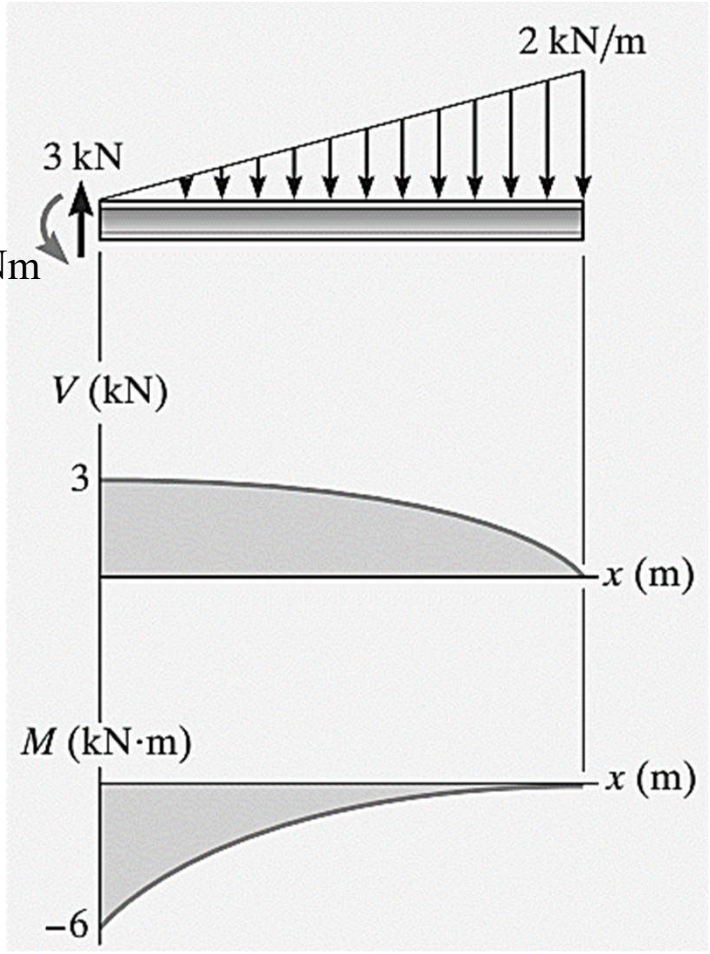
Shear Force and Bending Moments



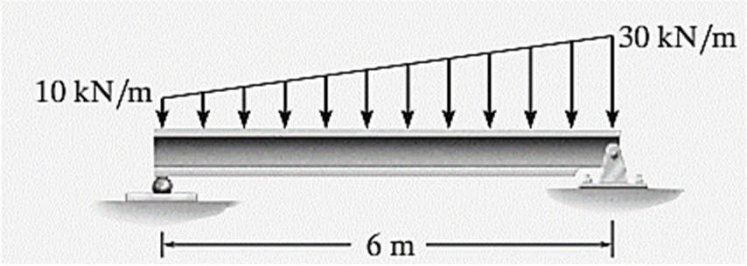
For segment ($0<x<3$)

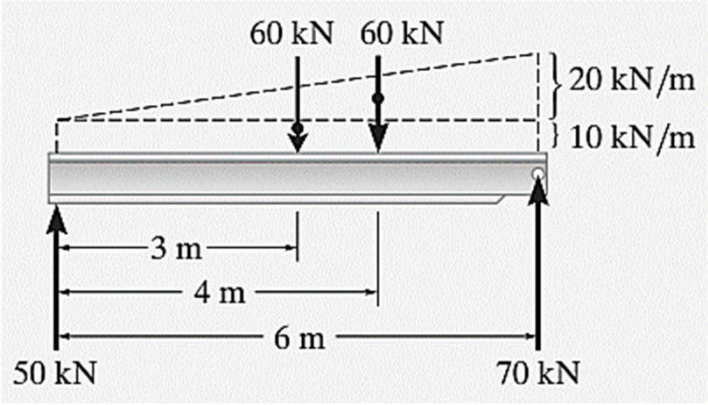
$$V = (3 - \frac{1}{3}x^2)\ \text{kN}$$

$$M = (-6 + 3x - \frac{1}{9}x^3)\ \text{kN}\cdot\text{m}$$



Shear Force and Bending Moments





For segment ($0<x<6$)

$$V = (50 - 10x - \frac{5}{3}x^2)\ \text{kN}$$

$$M = (50x - 5x^2 - \frac{5}{9}x^3)\ \text{kN}\cdot\text{m}$$

